Advanced Topics in Machine Learning

General information

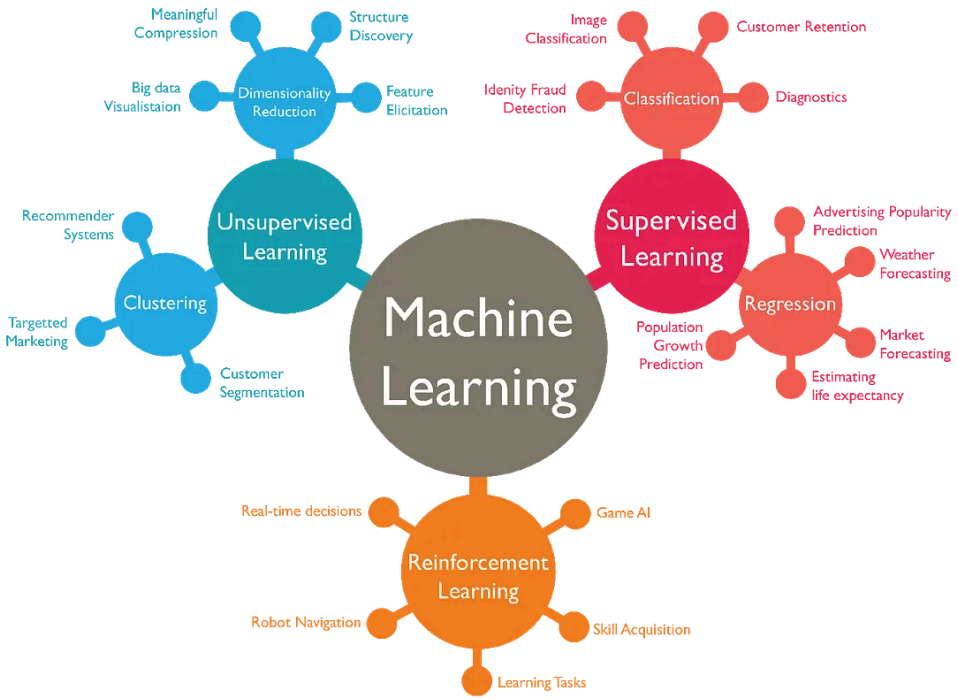
⚠️ All infos here are preliminiary and subject to change! ⚠️

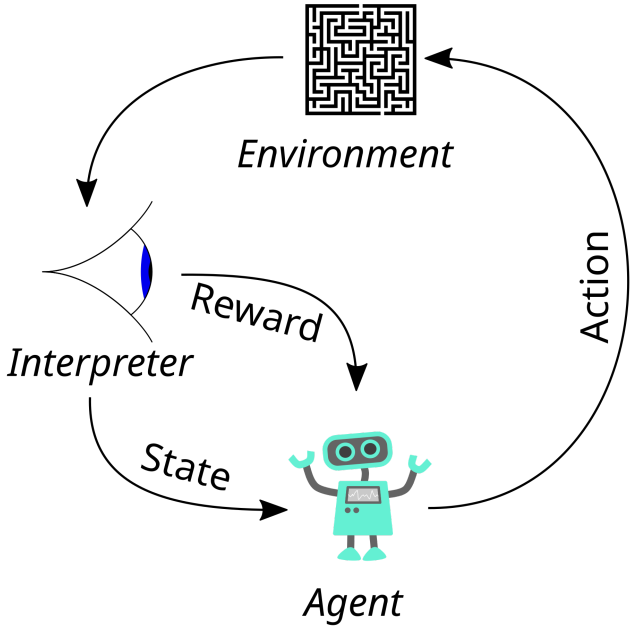
Official information will be given in the first lecture!

Course contents - First half

Reinforcement Learning (RL)

- Mathematical foundations:
  Markov decision processes (MDP)
- "Ideal" solutions:
  Dynamic Programming (DP)
- Model-free solutions:
  Monte-Carlo, Temporal Difference Learning

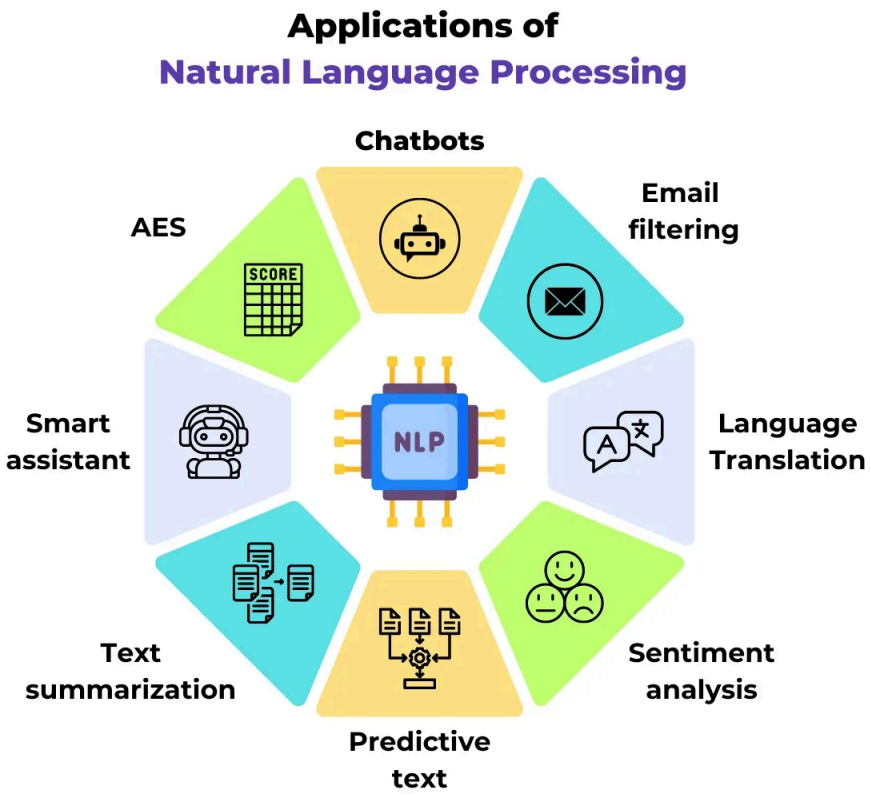




Course contents - Second half

Natural Language Processing (NLP)

- Tokenization, embeddings, ...
  ("How to feed text into an ML model?")
- Basic language models, recurrent neural networks, ...
- Transformers, attention, ...



People

Prof. Sebastian Engelke

- Lectures, projects, exams

Manuel Hentschel (TA)

- Seminars in the first half
- Projects/exams regarding **RL**

Marco Froelich (TA)

- Seminars in the second half
- Projects/exams regarding **NLP**

Evaluation

⚠️ *Preliminary information!* ⚠️ *Official infos in the lecture!* ⚠️

Reinforcement Learning:

- Individual project (ca. 20%)
- Written exam (ca. 30%)
- During the semester

Natural Language Processing:

- Group project+presentation, individual oral exam (ca. 50%)
- During the exam session

Organization

- Lectures: Tuesdays, 16:15-18:00, M4020
- Seminars: Thursdays, 16:15-18:00, M2140
- On Moodle:
  - General information, announcements, ...
  - Lecture slides
  - Project descriptions and submissions
- On GitHub:
  - Exercises, examples

The End